Convex Optimization Problems Cont' and Examples of Optimization

Lecture 7 for 18660/18460: Optimization

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Feb 3, 2026

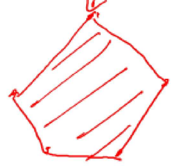
Admin Stuff

- HW 1 due Thursday (Feb 5)
- Reminder of late policies: 5 late days across semester, but 3 max for each homework
- Quiz for lecture 7 out, due Feb 5 before lecture

Recall

- Linear program

$$\min_{x\in\mathbb{R}^n} \; c^T x + d$$
$$Ax + b = 0$$
$$Gx + h \le 0$$



If optimizer exist one of them lies on "vertex"

- Quadratic program

$$\min_{x\in\mathbb{R}^n} \; \frac{1}{2}x^T P x + c^T x + d \quad (P \text{ is psd})$$
$$Ax + b = 0$$
$$Gx + h \le 0$$

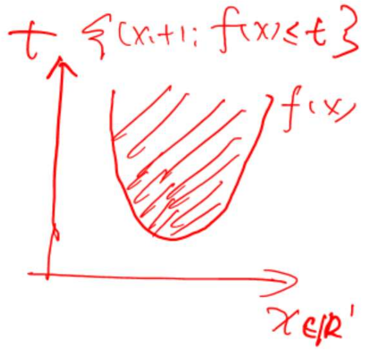
Epigraph Forms

(A)

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{s.t.} \quad x \in C$$

(B)

$$\min_{x\in\mathbb{R}^n, t\in\mathbb{R}} t \quad \text{s.t.} \quad x \in C, \quad f(x) \le t$$



Claim: if x^* is solution to (A), then $(x^*, \underbrace{f(x^*)}_{t^*})$ solves (B)

Pf: Assume (x^*, t^*) is NOT solution to (B)

Then there $\exists$ $(x', \underline{t'})$ s.t. $x' \in C$, AND $t' < t^*$

$f(x') \le t' < t^* = f(x^*)$ We find a better solution to (A) than x^* Contradiction!

Claim if (x^*, t^*) solves (B) $\Rightarrow$ x^* solves (A)

Epigraph Forms

$$f(x) = \max(c_1^\top x + d_1, c_2^\top x + d_2)$$

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{s.t.} \quad Gx \le h, \quad Ax = b$$

$$\min_{x\in\mathbb{R}^n, t\in\mathbb{R}} t \quad \text{s.t.} \quad Gx \le h, \quad Ax = b, \quad f(x) \le t$$

$\max(c_1^\top x + d_1, c_2^\top x + d_2) \le t$

$$\min_{x\in\mathbb{R}^n, t\in\mathbb{R}} t \quad \text{s.t.} \quad Gx \le h, \quad Ax = b, \quad c_1^\top x + d_1 \le t, \quad c_2^\top x + d_2 \le t$$

Change of Variables

If $\phi: \mathbb{R}^n \to \mathbb{R}^n$ is one-to-one, and its image covers feasible set C, then we can change variables in an optimization problem:

$x = \phi(z)$ ϕ needs to be "invertible"

$$\min_{x\in C} f(x) \quad \Leftrightarrow \quad \min_{z\in\phi^{-1}(C)} f(\phi(z))$$

$\{z : \phi(z) \in C\}$

- Could turn non-convex problems into convex problems!

$(e^{y_1})^2 (e^{y_2})^3$

$$\min_{x_1,x_2\in(0,+\infty)} x_1^2 x_2^3$$
$$\text{s.t. } x_1 x_2 + x_1 x_2^2 \le 1$$

This called Geometric Program

$x_1 = e^{y_1}$

$x_2 = e^{y_2}$

$y_1, y_2 \in \mathbb{R}$

$\Longrightarrow$ $\min_{y_1, y_2} e^{2y_1 + 3y_2}$

s.t. $e^{y_1+y_2} + e^{y_1+2y_2} \le 1$

convex: $e^{a^\top y}$ is convex

g is convex g(Ax+b) convex

Eliminating equality constraints

$$\min_x f(x)$$
$$g_i(x) \leq 0, i \in 1,2,\dots,k$$
$$Ax + b = 0$$

$$\min_z f(Fz + x_0)$$
$$g_i(Fz + x_0) \leq 0, i \in 1,2,\dots,k$$

where columns of F forms null space of A

$\{x : Ax + b = 0\} = \{Fz + x^0 : \}$

the "null space" of A

Any particular Solution

$A\hat{x} = b$

$\dim(z) = n - \text{rank}(A)$

$x_1 + x_2 = 1$

$x_1 = z$

$x_2 = 1 - z$

$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} z + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

free variable

$\min_{x_1, x_2, x_3} x_1^2 + x_2^2 + x_3^2$

s.t. $x_1 + x_2 + x_3 = 1$

$x_3 = 1 - x_1 - x_2$

$\min_{\underbrace{x_1 x_2}_{z}} x_1^2 + x_2^2 + (1 - x_1 - x_2)^2$

Quiz Results

(A) In convex optimization, a local minimizer must be a global minimizer.

(B) Suppose $f(x)$ is a differentiable function over $\mathbb{R}^n$. Suppose at a point x^*, $\nabla f(x^*) \neq 0$. Then, starting from x^* and moving along the negative gradient direction, i.e. $x^* - t\nabla f(x^*)$, there exists a $t^* > 0$ such that $f(x^* - t^*\nabla f(x^*)) < f(x^*)$

(C) Suppose $f(x)$ is a differentiable function. Suppose at a point x^*, $\nabla f(x^*) \neq 0$. Then, starting from x^* and moving along the negative gradient direction, i.e. $x^* - t\nabla f(x^*)$, we must have $f(x^* - t^*\nabla f(x^*)) < f(x^*)$ for all $t^* > 0$,

(D) The following problem can be transformed to LP

$\min \|x\|_\infty$

s.t. $Ax \geq b$

Outline for today

Today:
- Second-order cone programming
- Semi-definite programming

Optimization problems in engineering
- Energy and power systems: economic dispatch and optimal power flow

What are convex cones?

Convex sets

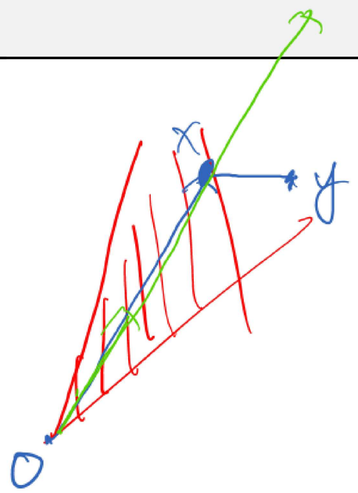
$\forall x, y \in C, \forall t \in [0,1], tx + (1-t)y \in C$

Convex cones

$\forall x, y \in C, \forall t \in [0,1], tx + (1-t)y \in C$

$\forall x \in C, \forall \alpha \in [0, \infty), \alpha x \in C$

$\alpha = 0 \Rightarrow 0 \in C$



Equivalent definition for convex cones

$\forall x, y \in C, \forall t_1, t_2 \in [0, \infty),$
$t_1 x + t_2 y \in C$

What are convex cones?

Convex sets	Convex cones
$\forall x, y \in C, \forall t \in [0,1], tx + (1-t)y \in C$	$\forall x, y \in C, \forall t \in [0,1], tx + (1-t)y \in C$ $\forall x \in C, \forall \alpha \in [0, \infty), \alpha x \in C$
Convex combination For $x_1, \ldots, x_m$, a convex combination is $t_1 x_1 + t_2 x_2 + \cdots t_m x_m$ For $t_i \geq 0, \forall i$, and $\sum_i t_i = 1$	**Conic combination** For $x_1, \ldots, x_m$, a conic combination is $t_1 x_1 + t_2 x_2 + \cdots t_m x_m$ For $t_i \geq 0, \forall i$
Convex hull Convex hull of set S is formed by convex combinations of all points in S	**Conic hull** Conic hull of set S is formed by conic combinations of all points in S

$x_1 \; x_2$

$t_1 x_1 + t_2 x_2$

$= t_1 x_1 + (1-t_1) x_2$

Convex Hull vs Conic Hull

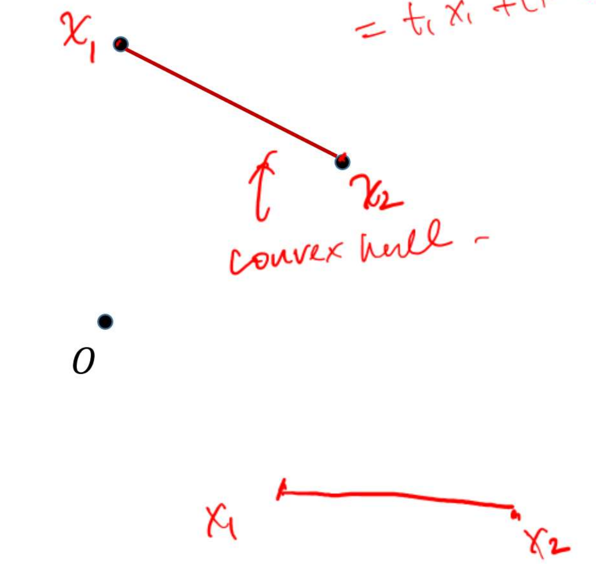
$S = \{x_1 \; x_2\}$

$t_1 \geq 0 , t_2 \geq 0$

$t_1 + t_2 = 1$

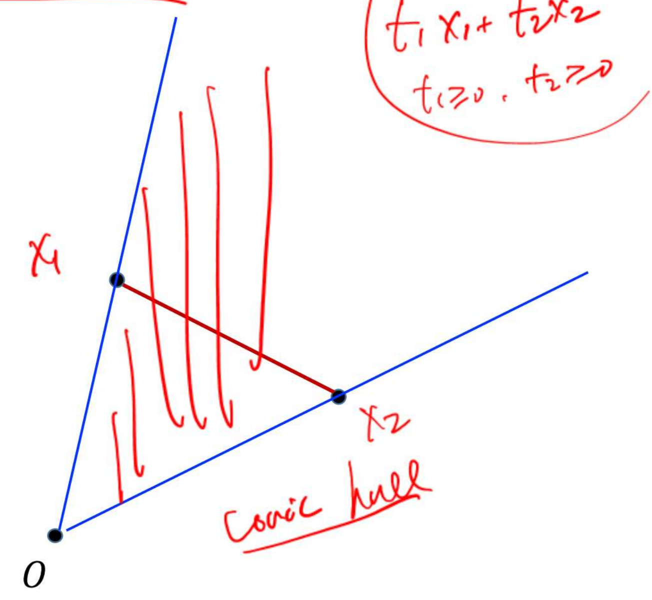
$t_1 x_1 + t_2 x_2$

$= t_1 x_1 + (1-t_1) x_2$



$t_1 x_1 + t_2 x_2$

$t_1 \geq 0 , t_2 \geq 0$



Norm Cones

$\in \mathbb{R}^n$ $t \in \mathbb{R}$

Definition. $\{[x, t]^\top \in \mathbb{R}^{n+1} : \|x\| \le t\}$, for a norm $\|\cdot\|$. Under ℓ_2 norm $\|\cdot\|_2$, this is called a second-order cone.

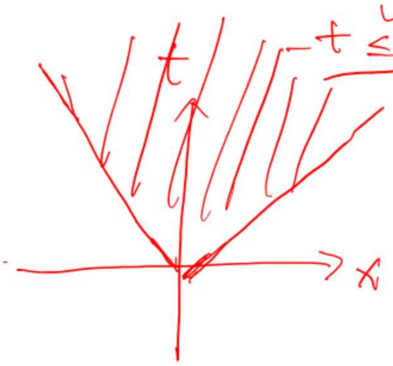
1-d

$n=1$ x scalar

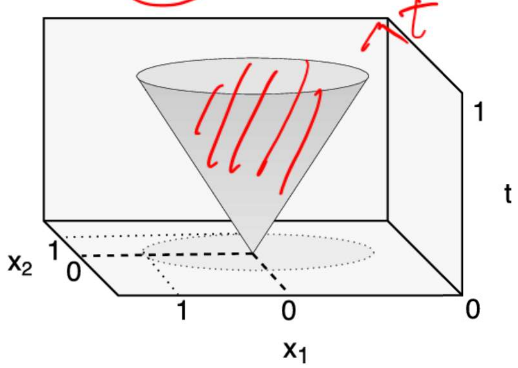
$\|x\|_2 \le t$

$\Leftrightarrow |x| \le t$

$-t \le x \le t$



x is 2-d



Second Order Cone Programming

$Ax \le b$

$x^T P x \le 1$

$\|P^{\frac{1}{2}} x\|_2^2 \le 1 \Leftrightarrow \|P^{\frac{1}{2}} x\|_2 \le 1$

$$\min \quad f^\top x$$

$$\|A_i x + b_i\|_2 \le c_i^\top x + d_i \quad i = 1,2,\dots,m$$

$$Fx = g$$

$(A_i \in \mathbb{R}^{n_i \times n}, b_i \in \mathbb{R}^{n_i})$

The inequalities are called second-order cone (SOC) constraints and can be written as

$$(A_i x + b_i, c_i^\top x + d_i) \in \text{second-order cone in } \mathbb{R}^{n_i+1}$$

Example: Robust LP

a_i is stock price for $a_i \in 1 \pm 0.05$

$a_i = 1 + 0.05 \cdot u$, $|u| \le 1$

- Consider LP with constraint $a_i^\top x + b_i \le 0$, but we have uncertainties about a_i and only knows $a_i \in \mathcal{E}_i = \{\bar{a}_i + P_i u \mid \| u \|_2 \le 1\}$,
- Want $a_i^\top x + b_i \le 0$ to hold for all possible $a_i \in \mathcal{E}_i$
- Consider constraint: $\max_{a_i \in \mathcal{E}_i} a_i^\top x + b_i \le 0$

$\in [0.95, 1.05]$

$a_i x \le 100$ $\quad a_i x - 100 \le 0$

$x = \frac{100}{1.05}$

$$
\begin{aligned}
a_i^\top x + b_i &= (\bar{a}_i + P_i u)^\top x + b_i \\
&= (\bar{a}_i)^\top x + \underbrace{u^\top P_i^\top x} + b_i \\
&\le (\bar{a}_i)^\top x + \|u\|_2 \cdot \|P_i^\top x\|_2 + b_i \\
&\le (\bar{a}_i)^\top x + \|P_i^\top x\|_2 + b_i
\end{aligned}
$$

≤ 0

Example: Robust LP

$x^2 - t \le 0$ $\quad t = \frac{(t+1)^2 - (t-1)^2}{4}$

$x^2 - \frac{(t+1)^2}{4} + \frac{(t-1)^2}{4} \le 0$

- Consider LP with constraint $a_i^\top x + b_i \le 0$, but we have uncertainties about a_i and only knows $a_i \in \mathcal{E}_i = \{\bar{a}_i + P_i u \mid \| u \|_2 \le 1\}$,
- Want $a_i^\top x + b_i \le 0$ to hold for all possible $a_i \in \mathcal{E}_i$
- Consider constraint: $\max_{a_i \in \mathcal{E}_i} a_i^\top x + b_i \le 0$

$x^2 + \frac{(t-1)^2}{4} \le \frac{(t+1)^2}{4}$

$\|(x, \frac{t-1}{2})\|_2^2 \le (\frac{t+1}{2})^2$

$\nabla^2 f(x) \succeq 0$

$\nabla^2 f(x) \succeq \mu I$

$h(\cancel{\ })$

$h(Ax+b)$

Robust LP

$$
\begin{aligned}
&\min c^\top x \\
\text{s.t.} \quad &\bar{a}_i^\top x + \|P_i^\top x\|_2 + b_i \le 0, \; i = 1, \dots, m
\end{aligned}
$$

$\|P_i^\top x\|_2 \le -\bar{a}_i^\top x - b_i$

$\|(x, \frac{t-1}{2})\|_2 \le \frac{t+1}{2}$

worst case $\quad a_i^\top x + b_i \le 0$

Outline for today

Today:
- Second-order cone programming
- **Semi-definite programming**

Optimization problems in engineering
- Energy and power systems: economic dispatch and optimal power flow

Semi-Definite Cones

$\mathbb{R}^n$

Definition. Suppose $\mathbb{S}^n$ is the set of n-by-n symmetric matrices. Semi-definite cone is the following subset $\mathbb{S}^n_+ = \{A \in \mathbb{S}^n : A \text{ is positive semi-definite}\}$.

Why this is convex cone?

A, B psd for $t \in [0,1]$: $tA + (1-t)B$ PSD

A psd. $\alpha \geq 0$, αA is psd

More on the $\preccurlyeq$ and $\succcurlyeq$ notation

- $\preccurlyeq$ and $\succcurlyeq$ defines a partial-order between symmetric matrices
- $A \succcurlyeq 0$ means A is positive semi-definite, all eigenvalues of A is ≥ 0
- $A \succcurlyeq B$ means $A - B \succcurlyeq 0$, i.e. $A - B$ is positive semidefinite $eig(A-B) \geq 0$
  - Caution! Eigenvalues of $A - B$ does NOT equal to $eig(A)$ – $eig(B)$
  - Exception: if $B = \mu I$, then eigenvalues of $A - B$ is $eig(A) - \mu$

- $A \preccurlyeq 0$ means $-A$ is positive semi-definite, all eigenvalues of A is ≤ 0
- $A \preccurlyeq B$ means $B - A \succcurlyeq 0$, i.e. $B - A$ is positive semidefinite

Semi-Definite Programming (SDP)

$F_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $F_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdots F_n$

$G = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$$\min_{x \in \mathbb{R}^n} c^\top x$$

s.t. $\quad x_1 F_1 + x_2 F_2 + \cdots + x_n F_n + G \preccurlyeq 0$

$\qquad Ax = b$

$$\underbrace{x_1 F_1}_{\begin{bmatrix} x_1 & 0 \\ 0 & 0 \end{bmatrix}} + \underbrace{x_2 F_2}_{\begin{bmatrix} x_2 & 0 \\ 0 & 0 \end{bmatrix}} + \cdots \underbrace{x_n F_n}_{\begin{bmatrix} x_n & 0 \\ 0 & 0 \end{bmatrix}} + \underbrace{G}_{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}$$

$$= \begin{bmatrix} x_1 + x_2 + \cdots + x_n + 1 & 0 \\ 0 & 0 \end{bmatrix} \preceq 0 \iff \begin{cases} x_1 + \cdots + x_n + 1 \leq 0 \\ 0 \leq 0 \end{cases}$$

Alternative Form of SDP

$a^\top x + b$

$= a_1 x_1 + a_2 x_2 + \cdots + a_n x_n + b$

- Treat entire matrix as a variable, $W \in \mathbb{S}^n$
- Can think of it as a vector of n^2 dimension
- Affine functions can still be defined on variable $W \in \mathbb{S}^n$, that is

$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ & & \vdots & \end{bmatrix}$

$W = \begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1n} \\ W_{21} & \cdots & & \end{bmatrix}$

$$\sum_{i,j} a_{ij} W_{ij} + b = tr(A^\top W) + b$$

- Where trace of a matrix can be defined as summation of diagonal entries

$$tr(X) = \sum_i X_{ii}$$

- Why the inner product $\sum_{i,j} a_{ij} W_{ij}$ can be written as $tr(A^\top W)$?

$[A^\top W]_{i,j} = \sum_k [A^\top]_{i,k} [W]_{kj} = \sum_k A_{ki} W_{kj}$

$tr(A^\top W) = \sum_{i=1}^n [A^\top W]_{i,i} = \sum_{i=1}^n \sum_{k=1}^n A_{ki} W_{ki}$

Alternative Form of SDP

$LP \subseteq QP \subseteq SOCP \subseteq SDP$

obvious

cone

Access Code.

$$\min_{W \in \mathbb{S}^n} \text{ affine func. on } W$$

s.t. $W \succcurlyeq 0$

affine equality constr. on W

$x^\top P x \le 1$

$\|P^{\frac{1}{2}} x\|_2 \le 1$

$A = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}$ $B = \begin{pmatrix} 0 & \\ & 1 \end{pmatrix}$

$A \not\succcurlyeq B$

$B \not\succcurlyeq A$

Alternative Form of SDP

$$\min_{W \in \mathbb{S}^n} \; tr(C^\top W)$$

$$\text{s.t.} \quad W \succcurlyeq 0$$

$$tr\left(A_i^\top W\right) = b_i, \text{ for } i = 1, \dots, m$$

$\Updownarrow$ Equivalent after an equivalent transformation

$$\min_{x \in \mathbb{R}^n} \; c^\top x$$

$$\text{s.t.} \quad x_1 F_1 + x_2 F_2 + \cdots + x_n F_n + G \preccurlyeq 0$$

$$Ax = b$$

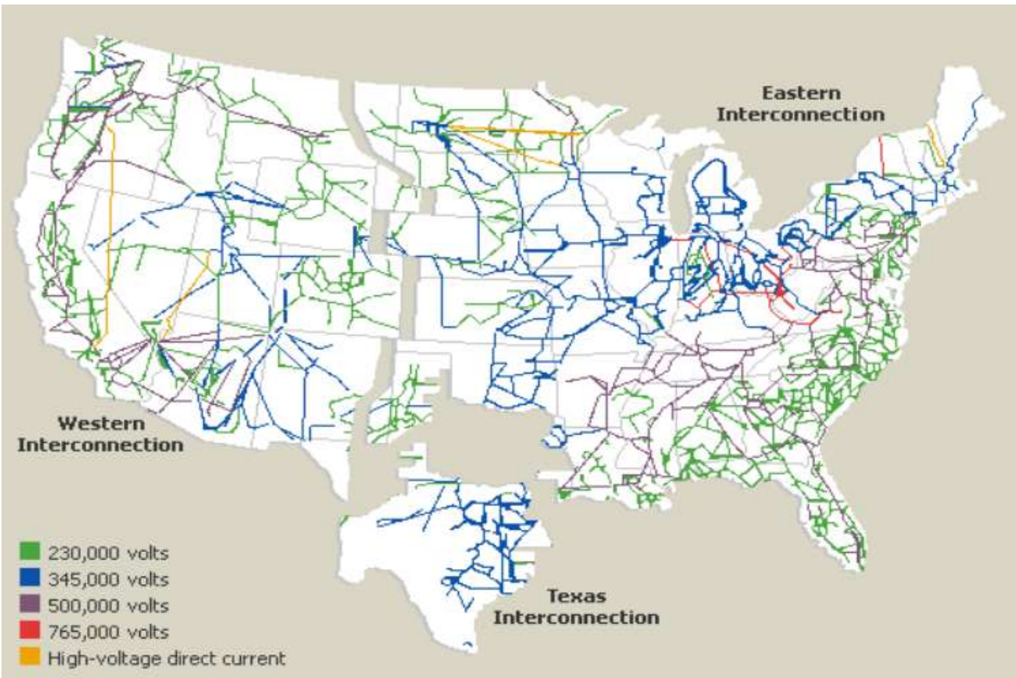
Summary so far

- Linear program
- Quadratic program
- Second-order cone program
- Semi-definite program
- Equivalent transforms
  - Epigraph forms
  - Eliminate affine equality constraints
  - Change of variables

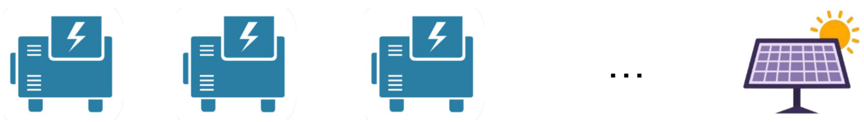
Up next: example in power systems

Power systems

Power systems



Economic Dispatch



m generators

generation	x_1	x_2	x_3		x_m
cost	$f_1(x_1)$	$f_2(x_2)$	$f_3(x_3)$		$f_m(x_m)$
limit	$[a_1, b_1]$	$[a_2, b_2]$	$[a_3, b_3]$		$[a_m, b_m]$
demand	d_1	d_2	d_3		d_n

n residents

Goal: match generation with demand with minimum cost!

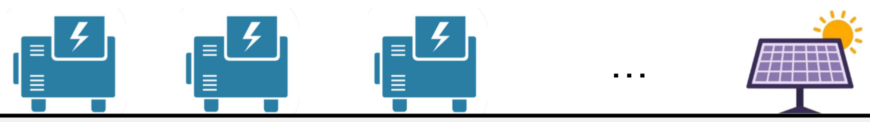
Economic Dispatch

$$\min_{x_1,\dots,x_n} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$\sum_{i=1}^{m} x_i = \sum_{j=1}^{n} d_j$$

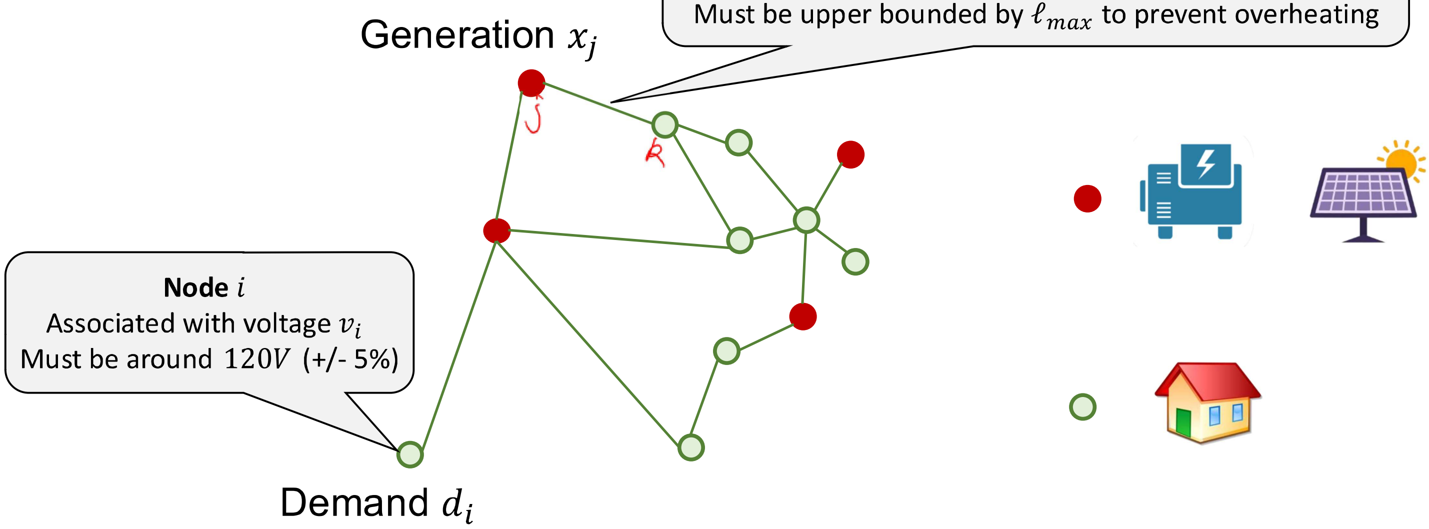
$$a_i \leq x_i \leq b_i$$

Economic Dispatch



This is actually simplistic as it does not consider the "network".

Optimal Power Flow



Generation, demand, voltage, current are related by power flow equations

$$PowerFlow(x, d, v, \ell) = 0$$

Circuits 101

$Voltage = Current * Resistance$

$Power = Voltage * Current$

$\Rightarrow$

$Power = (Voltage)^2/Resistance$

$Current = Voltage/Resistance$

$$p_j = \sum_{k:k\sim j} y_{jk}(v_j^2 - v_j v_k) = v^\top A_j v$$

Contains a quadratic equation!

$$\ell_{ij} = (v_i - v_j)y_{ij} = b_{ij}^\top v$$

Generation, demand, voltage, current are related by power flow equations

$$PowerFlow(x, d, v, \ell) = 0$$

Optimal Power Flow: Formulation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

Quadratic equality $\quad p_j = \sum_{k:k\sim j} y_{jk}(v_j^2 - v_j v_k) = v^\top A_j v, \forall j$

Affine equality $\quad \ell_{ij} = b_{ij}^\top v, \forall i, j$ connected

Affine inequality $\quad 120 * 0.95 \leq v_i \leq 120 * 1.05, \forall i$

Affine inequality $\quad \ell_{ij} \leq \ell_{max}$

What do we do about non-affine equality constraints?

Optimal Power Flow: Formulation

Idea 1: Linearization
- Simple, fast, used in industry
- Not accurate

Idea 2: Convex relaxation
- Much more accurate
- Very slow, not used in industry yet

What do we do about non-affine equality constraints?

Optimal Power Flow: Linearization

- Linearization is a generic idea to turn non-linear equations into linear equations
  - We will see this in the power example; will see the same in control/robotics

A non-affine equality	Approximated by	Affine equality
$h(x) = 0$	$\Rightarrow$	$h(x^{ref}) + \nabla h(x^{ref})^\top (x - x^{ref}) = 0$

Optimal Power Flow: Linearization

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = \sum_{k:k\sim j} y_{jk}(v_j^2 - v_j v_k) = v^\top A_j v, \forall j$$

Can be approximated by an affine constraint. Works reasonably well in practice.

$$\ell_{ij} = b_{ij}^\top v, \forall i,j \text{ connected}$$

$$120 * 0.95 \leq v_i \leq 120 * 1.05, \forall i$$

$$\ell_{ij} \leq \ell_{max}$$

Optimal Power Flow: Relaxation

Idea 1: Linearization
- Simple, fast, used in industry
- Not accurate

Idea 2: Convex relaxation
- Much more accurate
- Very slow, not used in industry yet

What do we do about non-affine equality constraints?

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = \sum_{k:k\sim j} y_{jk}(v_j^2 - v_j v_k) = v^\top A_j v, \forall j$$

$\ell_{ij} = b_{ij}^\top v, \forall i, j$ connected

$$120 * 0.95 \leq v_i \leq 120 * 1.05, \forall i$$

$$\ell_{ij} \leq \ell_{max}$$

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = \sum_{k:k\sim j} y_{jk}(v_j^2 - v_j v_k) = v^\top A_j v, \forall j$$

$$120 * 0.95 \leq v_i \leq 120 * 1.05, \forall i$$

Ignore the current constraint for simplicity

Optimal Power Flow: Relaxation

The trace trick

$$p_j = v^\top A_j v = tr(v^\top A_j v) = tr(A_j v v^\top) = tr(A_j W)$$

W and v related by

$$W = vv^\top$$

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = v^\top A_j v, \forall j$$

$$120 * 0.95 \le v_i \le 120 * 1.05, \forall i$$

$$\min_{x,v,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = tr(A_j W), \forall j$$

$$W = vv^\top$$

$$120 * 0.95 \le \underline{v_i} \le 120 * 1.05, \forall i$$

Can also represent using W

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = v^\top A_j v, \forall j$$

$$120 * 0.95 \leq v_i \leq 120 * 1.05, \forall i$$

$$\min_{x,v,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$

$$p_j = tr(A_j W), \forall j$$

$$W = vv^\top$$

$$(120 * 0.95)^2 \leq W_{ii} \leq (120 * 1.05)^2 \forall i$$

Optimal Power Flow: Relaxation

Lemma. W can be written as $W = vv^\top$ for some $v \in \mathbb{R}^n$ if and only if W is positive semi-definite and $rank(W) = 1$.

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = v^\top A_j v, \forall j$$
$$120 * 0.95 \le v_i \le 120 * 1.05, \forall i$$

$$\min_{x,v,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W = vv^\top$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

$$\min_{x,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W \succcurlyeq 0$$
$$rank(W) = 1$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = v^\top A_j v, \forall j$$
$$120 * 0.95 \le v_i \le 120 * 1.05, \forall i$$

$$\min_{x,v,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
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$$\min_{x,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W \succcurlyeq 0$$
$$rank(W) = 1$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

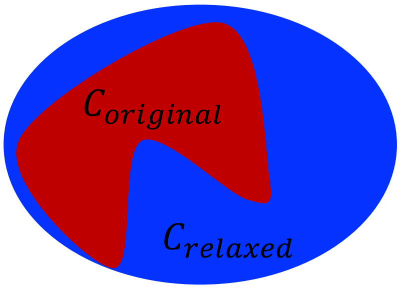
Non-equivalent transformation!

$$\min_{x,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W \succcurlyeq 0$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

Optimal Power Flow: Relaxation

$$\min_x f(x)$$
$$\text{s.t.} \quad x \in C_{original}$$

Original nonconvex problem



$$\min_x f(x)$$
$$\text{s.t.} \quad x \in C_{relaxed}$$

Relaxed convex problem

Lemma. If for the relaxed problem, the optimizer $x^*_{relaxed}$ lies in $C_{original}$, then $x^*_{relaxed}$ is also an optimizer for the original nonconvex problem. We call such scenarios as "the convex relaxation is tight."

Optimal Power Flow: Relaxation

$$\min_{x,v,\ell} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = v^\top A_j v, \forall j$$
$$120 * 0.95 \le v_i \le 120 * 1.05, \forall i$$

$\Leftrightarrow$

$$\min_{x,v,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W = vv^\top$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

$\Updownarrow$

$$\min_{x,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W \succcurlyeq 0$$
$$rank(W) = 1$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

$\Leftarrow$ Non-equivalent transformation!

$$\min_{x,\ell,W} f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$$
$$p_j = tr(A_j W), \forall j$$
$$W \succcurlyeq 0$$
$$(120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i$$

Optimal Power Flow: Relaxation

Theorem (Informal, [Low 2014]). For the optimal power flow problem below, the convex relaxation by dropping the rank constraint is tight under some conditions.

$$\begin{aligned} \min_{x,\ell,W} \ & f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \\ & p_j = tr(A_j W), \forall j \\ & W \succcurlyeq 0 \\ & (120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i \end{aligned}$$

Non-equivalent transformation!

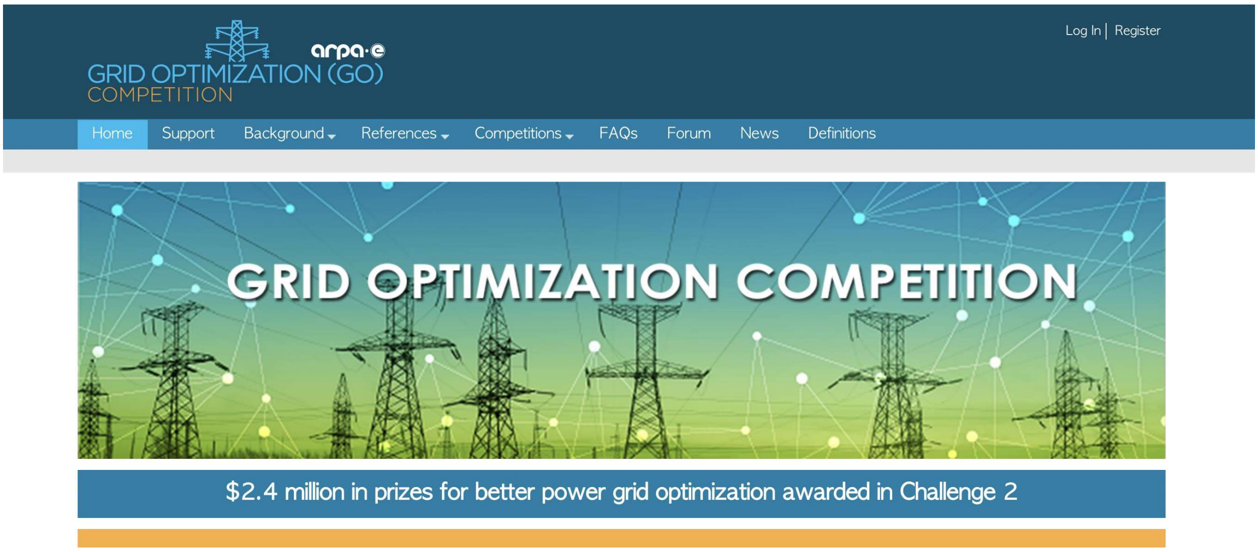
$$\begin{aligned} \min_{x,\ell,W} \ & f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \\ & p_j = tr(A_j W), \forall j \\ & W \succcurlyeq 0 \\ & rank(W) = 1 \\ & (120 * 0.95)^2 \le W_{ii} \le (120 * 1.05)^2 \forall i \end{aligned}$$

Optimal Power Flow: Relaxation

- Key trick: turn quadratic equation into matrix form, and then drop rank constraint

- Will see similar tricks in matrix completion (Netflix prize) next lecture.
- Pros: accurate, solves the original nonconvex program with quadratic equations
- Cons: semi-definite programs generally slow to solve; not used in industry yet

Department of Energy Sponsored OPF Competition



Summary for Optimal Power Flow

- Introduce decision variables
  - Natural decision: generation x_i
  - Additional variables involves in physics: voltage and current
- Write down cost: the generation cost
- Write down constraint
  - Physical law relating constraints (quadratic equality)
  - Operational constraint (voltage bounds, current bounds)
- Turn this into convex program
  - Linearization
  - Relaxation

Summary

Today:
- Second-order cone programming
- Semi-definite programming

Optimization problems in engineering
- Energy and power systems: economic dispatch and optimal power flow

Next lecture:
- Signal processing
- Machine learning
- Control/robotics